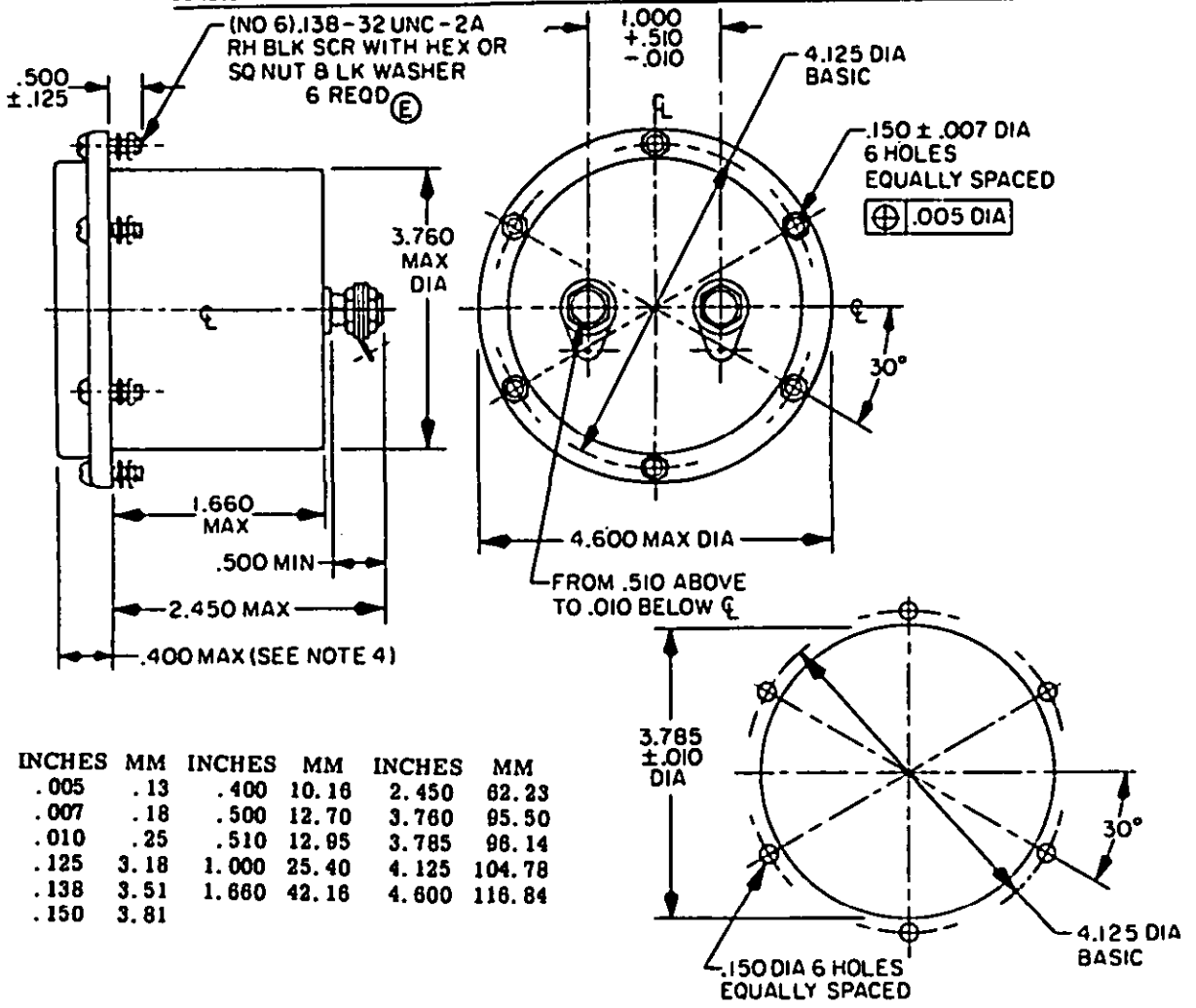
MIL-M-10304/17E
17 December 1975
SUPERSEDING
MIL-M-10304/17D
27 April 1971

MILITARY SPECIFICATION SHEET

METERS, ELECTRICAL INDICATING, PANEL TYPE, RUGGEDIZED: AMMETER, DC (FLUSH MOUNTING, ROUND FLANGE, 4-1/2 INCH) STYLE 46

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the meters described herein shall consist of this document and the latest issue of Specification MIL-M-10304.



PANEL DRILLING PLAN

INCHES	MM	INCHES	MM	INCHES	MM
.005	.13	.400	10.16	2.450	62.23
.007	.18	.500	12.70	3.760	95.50
.010	.25	.510	12.95	3.785	96.14
.125	3.18	1.000	25.40	4.125	104.78
.138	3.51	1.660	42.16	4.600	116.84
.150	3.81				

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Tolerance is ±1/2° on angles.
4. Maximum projection from front of panel including mounting gaskets (if used).

Ⓔ denotes changes

FSC 6625

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:50Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:
(Readings expressed as a percent of full-scale value.)

Dimensions and configuration: See figure.

Full-scale ranges: See table I.

Scale:
Length: 3.25 inches minimum.
Pointer deflection: 90° minimum.

Position influence: ±2 percent change for 60° rotation from normal operating position.

Accuracy: ±2 percent.

(E) Overshoot: See table I.

Response time: See table I.

Power consumption (loss) (at end scale deflection): See table.

High temperature cycling:
±4 percent.
±2 percent permanent change.

Temperature influence: ±1 percent change.
(Type MR46W001DCMAR meters shall be tested as millivoltmeters in this test.)

Exposure to extreme temperatures:
±3 percent permanent change.
(Type MR46W001DCMAR meters shall be tested as millivoltmeters in this test.)

Overload capacity:
Momentary overload: ±2 percent permanent change.

Sustained overload:
±1 percent temporary zero shift.
±1 percent permanent zero shift.
±2 percent permanent change.

Dielectric withstanding voltage: 3,000 volts rms.

Vibration, random drop, and shock: Repeatability is not applicable for those microammeters below 100-microampere ranges.

TABLE I. Type designation and characteristics.

Type designation	Full-scale ranges		Overshoot (max)	Resistance	Power consumption (loss)	Response time
	Zero	Range	%	ohms	millivolts	seconds max
MR46W001DCMAR	Left	1 mA 100mV	40	100 nom	100±2	2
MR46W050DCUAR	Left	50 µA	50	2,500±2%	125 nom	3

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:50Z
Check the source to verify that this is the current version before use.

Custodians:
Army - EL
Navy - EC
Air Force - 82

Review activities:
Army - EL, AV, AT, ME, WC
Navy - EC, MC
Air Force - 11
DSA - ES

User activities:
Army - MI
Navy - CG
Air Force - None

Preparing activity:
Army - EL

Agent:
DSA - ES

(Project 6625-0574-13)

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:50Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-M-10304/17E METERS, ELECTRICAL INDICATING, PANEL TYPE, RUGGEDIZED: AMMETER, DC (FLUSH MOUNTING, ROUND FLANGE, 4½ INCH) STYLE 46

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

To detach this form, along this line

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:50Z
Check the source to verify that this is the current version before use.